

MonetDB has a three-layer architecture. The top layer is for providing an SQL interface, the middle layer holds optimisers for MonetDB Assembly Language (MAL), and the bottom-most layer is the database kernel for providing access to the Binary Association Tables.

Installation

Detailed installation instructions for various platforms are available in the official documentation site, <https://www.monetdb.org/Downloads>. Though Windows 32-bit and 64-bit installers are given, the official documentation recommends Linux 64-bit binaries for scenarios in which the tables are expected to grow larger than 2GB disk space. If you wish to compile from the source, follow the instructions at <https://www.monetdb.org/Developers/SourceCompile>.

Getting started

In the Linux environment, start the MonetDB daemon *monetdbd*. This daemon is controlled by the *monetdb* application. A simple session demonstration is shown in the following code segment:

```
shell> monetdbd create /path/to/mydbfarm
shell> monetdbd start /path/to/mydbfarm
shell> monetdb create voc
shell> monetdb release voc
shell> mclient -u monetdb -d voc
password:<monetdb>
Welcome to mclient, the MonetDB/SQL interactive terminal
(Oct2014-SP1)
Database: MonetDB v11.19.7 (Oct2014-SP1), 'map:monetdb://localhost:50000/voc'
Type \q to quit, \? for a list of available commands
auto commit mode: on
sql>SELECT 'hello world';
+-----+
| single_value |
+-----+
| hello world  |
+-----+
1 tuple (0.530ms)
```

The following code uses the VOC data set, which is available at https://www.monetdb.org/sites/default/files/voc_dump.zip. It's a compressed file consisting of SQL statements. After the extraction of its contents, it is loaded into MonetDB with the help of *mclient*. The official documentation suggests that before loading the VOC data set, it is better to create a new user. The user creation and schema setting is shown below:

```
shell> mclient -u monetdb -d voc
password:<monetdb>
```

```
sql>CREATE USER "voc" WITH PASSWORD 'voc' NAME 'VOC Explorer'
SCHEMA "sys";
sql>CREATE SCHEMA "voc" AUTHORIZATION "voc";
sql>ALTER USER "voc" SET SCHEMA "voc";
```

A test table is created as shown below:

```
sql>CREATE TABLE test (id int, data varchar(30));
operation successful (0.722ms)
```

voc_dump.sql can be imported with the use of the following code:

```
shell> mclient -u voc -d voc voc_dump.sql
password:<voc>
```

The VOC data set has data for about 8000 voyages. You may check the data that you loaded using the steps mentioned earlier, with the following *SELECT* statement.

```
sql>SELECT count(*) FROM voyages;
+-----+
| L1    |
+-----+
| 8131  |
+-----+
```

Various operations can be carried out with the loaded data set. A sample is given below:

```
sql>SELECT chamber, CAST(AVG(invoice) AS integer) AS average
FROM invoices
WHERE invoice IS NOT NULL
GROUP BY chamber
ORDER BY average DESC;
+-----+-----+
| chamber | average |
+-----+-----+
| A       | 282996  |
| Z       | 259300  |
| H       | 150182  |
| R       | 149628  |
| D       | 149522  |
| E       | 149518  |
| null    | 83309   |
+-----+-----+
```

Detailed instructions on all the SQL operations are provided in the official documentation.

Extensions

MonetDB has support for various extensions.

- **GeoSpatial:** MonetDB has an interface to the